ESTIMATING IN BUILDING CONSTRUCTION

Ch. 1 Introduction to Estimating

(Construction Cost Estimate and Control)

Assistant Professor

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Introduction

- Class Introduction
- Self Introduction

Estimating in Building Construction

- The determination of probable construction costs for a project
- Prepared from the contract documents:
  - Plans
  - Project manual (plans, contract documents and technical specifications).

Estimating in Building Construction

- Quantities of estimates are used to:
  - Establish project budget
  - Order materials

Profit Margin (Engineering Economics)

- Provides the general contractor an acceptable rate of return and compensation for the risk associated with the project

Types of Estimates

- Detailed estimate
  - Determines the quantity and cost of everything required for the project, including:
    - Materials
    - Labor
    - Equipment
    - Insurance
    - Bonds
    - Overhead
    - Profits, Etc.

Types of Estimates

- Assembly estimating
  - Bid by groups of components (assemblies)
  - For example, a light switch would include:
    - Single-gang box
    - Single-pole switch
    - Cover plate
    - Two wire nuts
    - 20 feet of wire

Types of Estimates

- Square-foot estimates
  - Bid based on an average cost per square foot
  - May be adjusted for:
    - Building height
    - Building perimeter, Etc.
  - Sometimes, a unit other than square footage is used. E.g. Parking garage are measured by the number of parking stalls in the garage.

Types of Estimates

- Parametric estimates
  - Parametric estimates use equations that express the statistical relationship between building parameters and the cost of the building. Bid based upon the statistical relationship between building components may include:
    - Square footage
    - Number of floors
    - Length of perimeter
    - Percentage of the building that is common space
    - Etc.

Regression Equations

Types of Estimates

- Model estimating
  - Prepares an estimate based upon the answers to a number of questions
  - Questions for the slab on grade portion of the building might include:
    - What is the thickness of the slab?
    - Does the slab have wire mesh?
    - Do you want fire sprinklers?

Types of Estimates

- Project comparison
  - Prepared by comparing the costs of a proposed project to a similar completed project
  - Adjusted for difference between the comparison project and the project being estimated

Estimating Opportunities

- Architectural offices
- Engineering offices
- Subcontractors
- Material suppliers
- Manufacturers' representatives
- Project management
- Government
- Professional quantity surveyor
- Freelance estimators
- Residential construction
- Computer Software

Quantity Takeoff Skills

- Read and quantify plans
- Knowledge of math and geometry
- Patience
- Does careful, thorough work
- Computer literate with takeoff software
  - On-Screen Takeoff
  - Paydirt for cut and fill

Additional Estimating Skills

- Visualize building from plans
- Foresee problems
- Experienced in the construction process
  - Methods of handling materials
  - Labor productivity
- Convert labor productivity to costs
- Maintain a cost database

Additional Estimating Skills

- Computer literate with:
  - Spreadsheets
  - Estimating software such as WinEst
- Meets bid deadlines
- Strong writing and presentation skills

Types of Bids

- Competitive
  - Awarded based upon:
    - Price
    - Best value
- Negotiated bids
  - Owner and contractor agree upon a price

Contract Documents

- Agreement
- General conditions
- Supplementary general conditions
- Working drawings
- Specifications
- Addenda

Bid Information

- Publicly advertized in:
  - Newspapers
  - Trade journals
    - Engineering New Record (ENR)
    - Associated General Contractors
  - Internet

Sources of Information

- Historical data
  - Best source
- Construction cost guides
  - Averages for a few contractors

Thank You